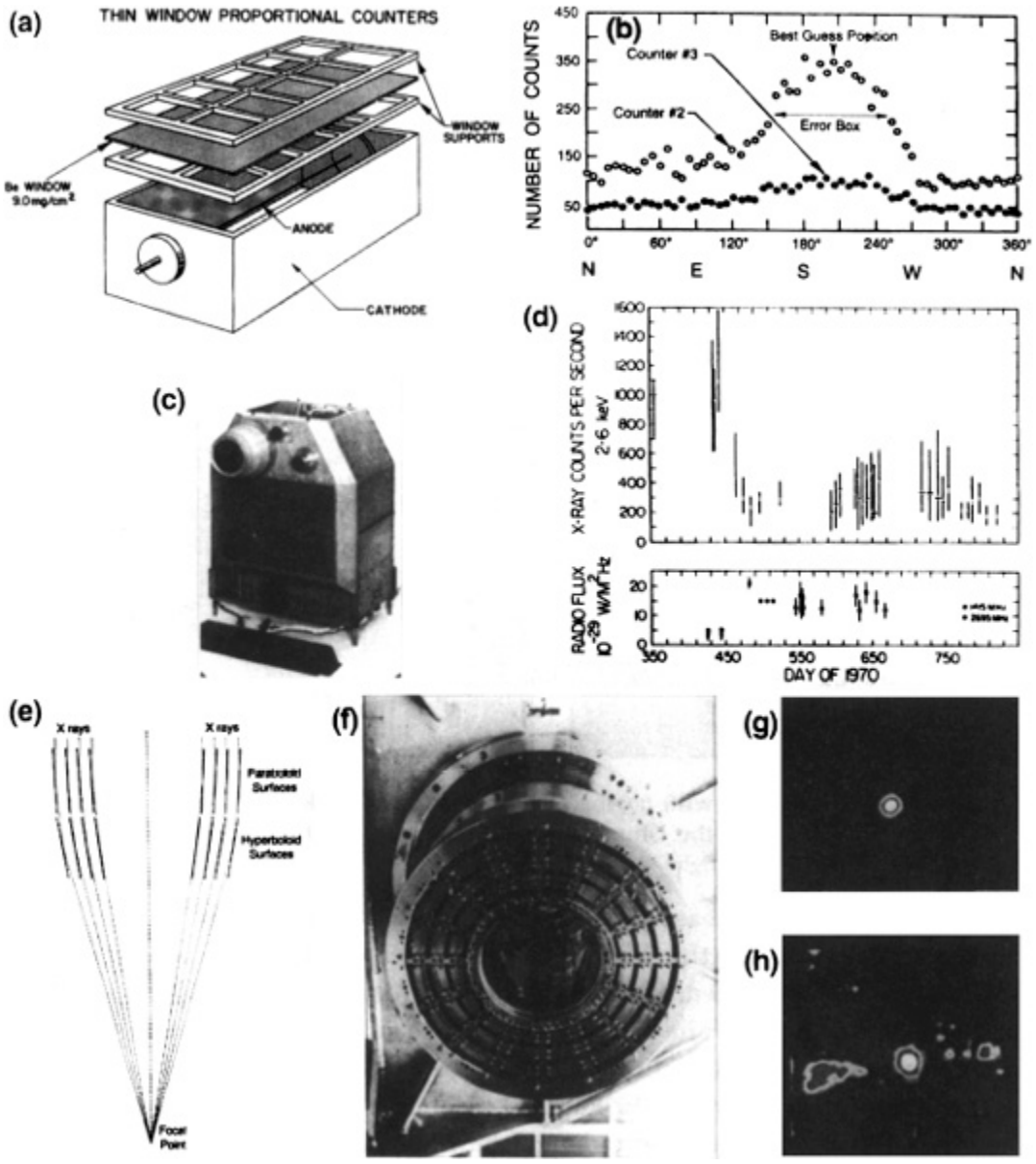


8.6 The improving technology and performance of X-ray astronomy's tools, 1962–1978. (a) Schematic design of the Geiger counter used by Giacconi's team in their 1962 discovery of the first X-ray star. (b) The data from that Geiger counter, showing that the star was not at the location of the Moon; note the very poor angular resolution (large error box), 90 degrees. (c) The 1970 Uhuru X-ray detector: A vastly improved Geiger counter sits inside the box, and in front of the counter one sees venetian-blind slats that prevent the counter from detecting an X-ray unless it arrives nearly perpendicular to the counter's window. (d) Uhuru's measurements of X-rays from the black-hole candidate Cygnus X-t. (e) Schematic diagram and (I) photograph of the mirrors that focus X-rays in the 1978 X-ray telescope Einstein. (g, h) Photographs made by the Einstein telescope of two black-hole candidates, Cygnus X-1 and SS-433. [Individual drawings and pictures courtesy R. Giacconi.]

How had the theorists gone wrong? How had they underestimated by a factor of 5000 the strengths of cosmic X-rays? They had presumed,